

D581

Introduction to Research Methods

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A researcher at a nonprofit with a small grant from a family foundation wants to understand how first-generation college students describe their sense of belonging during their first semester. Which approach fits best?

- A. A quantitative design, because all research must produce numbers.
- B. A randomized experiment comparing two belonging programs.
- C. A closed-ended survey testing whether belonging predicts GPA.
- D. A qualitative design using in-depth interviews and thematic analysis.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A study at a university library that recently extended its hours compares reading-test scores of students who study with background music and students who study in silence. Which statement is the null hypothesis?

- A. There is no difference in mean reading scores between the music and silence groups.
- B. Music will lower the score of every student.
- C. Extending library hours raises reading scores.
- D. Students who study with music will score lower than students who study in silence.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

Which statement is a testable research hypothesis?

- A. Safety training always eliminates workplace injuries.
- B. Safety training is a good thing.
- C. Injuries happen because of bad luck.
- D. Employees who complete a four-week safety course will report fewer injuries in six months than employees who do not.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

In a study at a 400-bed hospital, nurses are assigned to either 8-hour or 12-hour shifts, and medication error rates are compared between the groups. The study is funded by a nursing association. What is the dependent variable?

- A. The nurses.
- B. The size of the hospital.
- C. The medication error rate.
- D. Shift length.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A study finds that students who sleep more earn higher test scores. Researchers then note that students who sleep more also work fewer hours at part-time jobs, which could also affect scores. In this study, what is part-time work?

- A. The dependent variable.
- B. The independent variable.
- C. A confounding variable.
- D. A control variable held constant by design.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A school district with 12 elementary schools and a new superintendent is evaluating a reading program. Principals chose whether to adopt it, so 6 schools use it and 6 do not, and year-end reading scores are compared. What kind of design is this?

- A. Not research at all, because the schools chose.
- B. Quasi-experimental.
- C. A true experiment.
- D. A qualitative case study.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A researcher randomly selects 500 adults from a statewide voter list, which was updated in January, and then lets each person choose whether to join a walking program. Which statement best describes this study?

- A. The January update makes the sample invalid.
- B. Results may generalize to the state's adults, but self-chosen groups weaken any claim that walking caused the outcomes.
- C. It supports causal claims but cannot generalize, since people chose their groups.
- D. It supports strong causal claims because participants were chosen at random.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A lab randomly assigns 60 volunteer students from one psychology class to drink caffeine or a placebo before a memory test. What is the main strength and limit?

- A. Strong causal conclusions, but limited generalization beyond these volunteers.
- B. Strong generalization, but weak causal conclusions.
- C. Useless unless it has at least 600 students.
- D. Strong on both, because randomness was used.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A city evaluates a 12-month quit-smoking program by comparing smoking rates before and after. During that same year, the state raised cigarette taxes by \$1.50 a pack. The program's coordinator has 20 years of experience. Smoking fell 18%. What is the main threat to concluding the program caused the drop?

- A. The coordinator's experience inflated the results.
- B. None; a before-after design always shows the program's effect.
- C. History: the tax increase could explain the drop.
- D. Maturation: participants naturally aged.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Using a library database the university just licensed, a student finds a news story about a new trial, a textbook chapter summarizing research on the topic, and a journal article in which the authors report the trial they ran with 300 patients. Which is a primary source?

- A.** The textbook chapter.
- B.** The journal article reporting the authors' own trial.
- C.** All three, since they discuss the same trial.
- D.** The news story.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A student searching a database that indexes 2 million articles gets 45,000 results for exercise. She wants only articles about exercise and depression in adolescents. Which search is best?

- A.** exercise OR depression OR adolescents
- B.** exercise NOT depression
- C.** exercise AND depression
- D.** exercise AND depression AND adolescents

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A search for teenagers AND anxiety returns only 12 results. Which search will broaden results to catch other words for the same age group?

- A.** (teenagers OR adolescents OR youth) AND anxiety
- B.** teenagers AND adolescents AND youth AND anxiety
- C.** teenagers AND anxiety AND treatment
- D.** teenagers NOT anxiety

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A researcher at a nonprofit ~~with a small grant from a family foundation~~ wants to understand how first-generation college students describe their sense of belonging during their first semester. Which approach fits best?

Target word: fitting methodology **Decides:** The question asks how students describe an experience.

A	Absolute language	Method follows the question; not all research is numeric.
B	Adjacent category	Tests an effect; the question asks for descriptions, not effects.
C	Confusion set	Answers a different, quantitative question.
D	Correct	Exploring meaning and experience in participants' own words is qualitative work.

2. Answer A Concept 2

A study at a university library ~~that recently extended its hours~~ compares reading-test scores of students who study with background music and students who study in silence. Which statement is the null hypothesis?

Target word: null hypothesis **Decides:** The null states no difference.

A	Correct	The null asserts no effect or no difference.
B	Absolute language	Hypotheses concern group differences, not every individual.
C	Camouflage	Library hours are not the variable under study.
D	Confusion set	That is a directional alternative hypothesis.

3. Answer D Concept 2 · transfer

Which statement is a testable research hypothesis?

Target word: falsifiable hypothesis **Decides:** Names variables, groups, and a measurable expected result.

A	Absolute language	An absolute claim, not a realistic test of difference.
B	Dropped element	No measurable outcome or comparison; a value statement.
C	Confusion set	Unmeasurable cause; data cannot contradict it.
D	Correct	Specific, measurable, and could be proved wrong by data.

4. Answer C Concept 3

In a study ~~at a 400-bed hospital~~, nurses are assigned to either 8-hour or 12-hour shifts, and medication error rates are compared between the groups. ~~The study is funded by a nursing association.~~ What is the dependent variable?

Target word: dependent variable **Decides:** The outcome measured in each group.

A	Adjacent category	The nurses are participants, not a variable.
B	Camouflage	Bed count describes the setting, not a study variable.
C	Correct	The measured outcome is the dependent variable.
D	Inverted operation	Shift length is varied by the researchers; it is the independent variable.

5. Answer C Concept 3 • transfer

A study finds that students who sleep more earn higher test scores. Researchers then note that students who sleep more also work fewer hours at part-time jobs, which could also affect scores. In this study, what is part-time work?

Target word: role of the third factor **Decides:** An uncontrolled factor linked to both sleep and scores.

A	Confusion set	Test scores are the outcome.
B	Adjacent category	Sleep is the variable of interest; work hours were not studied on purpose.
C	Correct	It varies with the IV and could explain the outcome.
D	Dropped element	It was not held constant; that is why it threatens the conclusion.

6. Answer B Concept 4

A school district with 12 elementary schools ~~and a new superintendent~~ is evaluating a reading program. Principals chose whether to adopt it, so 6 schools use it and 6 do not, and year-end reading scores are compared. What kind of design is this?

Target word: design type **Decides:** Groups were formed by choice, not randomly.

A	Absolute language	Non-random groups weaken causal claims but still form a legitimate design.
B	Correct	Comparison groups exist, but assignment was not random.
C	Dropped element	Leaves out the requirement of random assignment.
D	Adjacent category	Scores are compared numerically across groups.

7. Answer B Concept 5

A researcher randomly selects 500 adults from a statewide voter list, ~~which was updated in January~~, and then lets each person choose whether to join a walking program. Which statement best describes this study?

Target word: what random selection gives and what is missing **Decides:** Random selection but self-selected groups.

A	Camouflage	The list date does not decide the design's strengths.
B	Correct	Random sampling aids external validity; lacking random assignment hurts internal validity.
C	Inverted operation	Reverses which validity each feature affects.
D	Confusion set	Random selection is not random assignment.

8. Answer A Concept 5 • transfer

A lab randomly assigns 60 volunteer students from one psychology class to drink caffeine or a placebo before a memory test. What is the main strength and limit?

Target word: strength and limit **Decides:** Random assignment but a convenience sample.

A	Correct	Random assignment supports cause; a non-random sample limits external validity.
B	Inverted operation	Reverses the effects of assignment and sampling.
C	Wrong stem number	Sample size affects precision, not the logic of assignment vs. sampling.
D	Confusion set	Randomness in assignment does not make the sample representative.

9. Answer C Concept 6

A city evaluates a 12-month quit-smoking program by comparing smoking rates before and after. During that same year, the state raised cigarette taxes by \$1.50 a pack. ~~The program's coordinator has 20 years of experience.~~ Smoking fell 18%. What is the main threat to concluding the program caused the drop?

Target word: validity threat **Decides:** An outside event occurred during the study.

A	Camouflage	Experience does not create an alternative explanation for the drop.
B	Absolute language	Pre/post designs without a comparison group are open to many threats.
C	Correct	An outside event during the study period is a history threat.
D	Adjacent category	Maturation is change within people over time, not an outside event.

10. Answer B Concept 7

~~Using a library database the university just licensed,~~ a student finds a news story about a new trial, a textbook chapter summarizing research on the topic, and a journal article in which the authors report the trial they ran with 300 patients. Which is a primary source?

Target word: primary source **Decides:** The authors report their own original data.

A	Confusion set	Summarizes others' work, so it is secondary.
B	Correct	Original research reported by those who did it is primary.
C	Absolute language	Topic does not make a source primary; authorship of the data does.
D	Adjacent category	A popular secondary account.

11. Answer D Concept 8

A student searching a database ~~that indexes 2 million articles~~ gets 45,000 results for exercise. She wants only articles about exercise and depression in adolescents. Which search is best?

Target word: narrowing search **Decides:** All three terms must be present.

A	Inverted operation	OR broadens; results will grow.
B	Confusion set	Excludes the very topic she wants.
C	Dropped element	Leaves out the adolescent term.
D	Correct	AND requires every term, narrowing results.

12. Answer A Concept 8 · transfer

A search for teenagers AND anxiety returns only 12 results. Which search will broaden results to catch other words for the same age group?

Target word: broadening with synonyms **Decides:** OR among synonyms, AND with the topic.

A	Correct	OR accepts any synonym while AND keeps the topic.
B	Inverted operation	Requires all synonyms at once, narrowing further.
C	Adjacent category	Adding a term with AND narrows.
D	Confusion set	Excludes the topic.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	5	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Question drives method	Questions about how people experience or make meaning of something call for qualitative methods; questions about how much or whether X affects Y call for quantitative designs.	Assuming numbers always make research stronger.	Words like describe, experience, or meaning vs. effect, difference, or relationship.
2	Testable hypotheses	A hypothesis must be specific and falsifiable, naming variables and an expected result that data could contradict; the null states no effect or no difference.	Writing a value statement or an absolute claim as a hypothesis.	Choosing which statement could be tested.
3	Independent, dependent, and confounding variables	The independent variable is what is varied or compared; the dependent variable is the measured outcome; a confound is an uncontrolled factor tied to both.	Swapping IV and DV, or calling a confound the IV.	"What is the dependent variable?" or a third factor moving with the IV.
4	Experimental vs. quasi-experimental designs	A true experiment requires random assignment to conditions; comparison groups without random assignment make a quasi-experiment.	Calling any study with a comparison group an experiment.	Who decided which group received the treatment.
5	Random sampling vs. random assignment	Random sampling supports generalizing to a population (external validity); random assignment supports causal conclusions (internal validity).	Thinking a random sample alone lets you claim cause.	How participants were chosen vs. how they were placed into groups.
6	Threats to internal validity	History, maturation, selection, testing, and attrition can explain pre/post changes; an outside event during the study is a history threat.	Crediting the program for any before-after change.	An event outside the program happened during the study period.
7	Primary vs. secondary sources	A primary source reports original research by the people who did it; secondary sources summarize or interpret others' work.	Treating textbooks, reviews, or news summaries as primary.	Whether the authors collected the data themselves.
8	Boolean search logic	AND narrows by requiring all terms; OR broadens by accepting any synonym; NOT excludes a term.	Using OR to narrow or AND to capture synonyms.	Too many or too few results in a database search.

Course scope (public description): Introduction to Research Methods familiarizes students with the foundations of research, guiding students through selecting topics, forming research questions, engaging with relevant and reliable literature, and designing a research project. This course provides an overview of how researchers form questions and hypotheses, which different types of methodologies can be used to address these questions, and how existing literature and data are used to support or reject hypotheses. The methods presented in this course can be applied to many fields and disciplines, and they provide a baseline for students to use in their chosen area

of study and future work.